

Community Emergency Response App (CERA)

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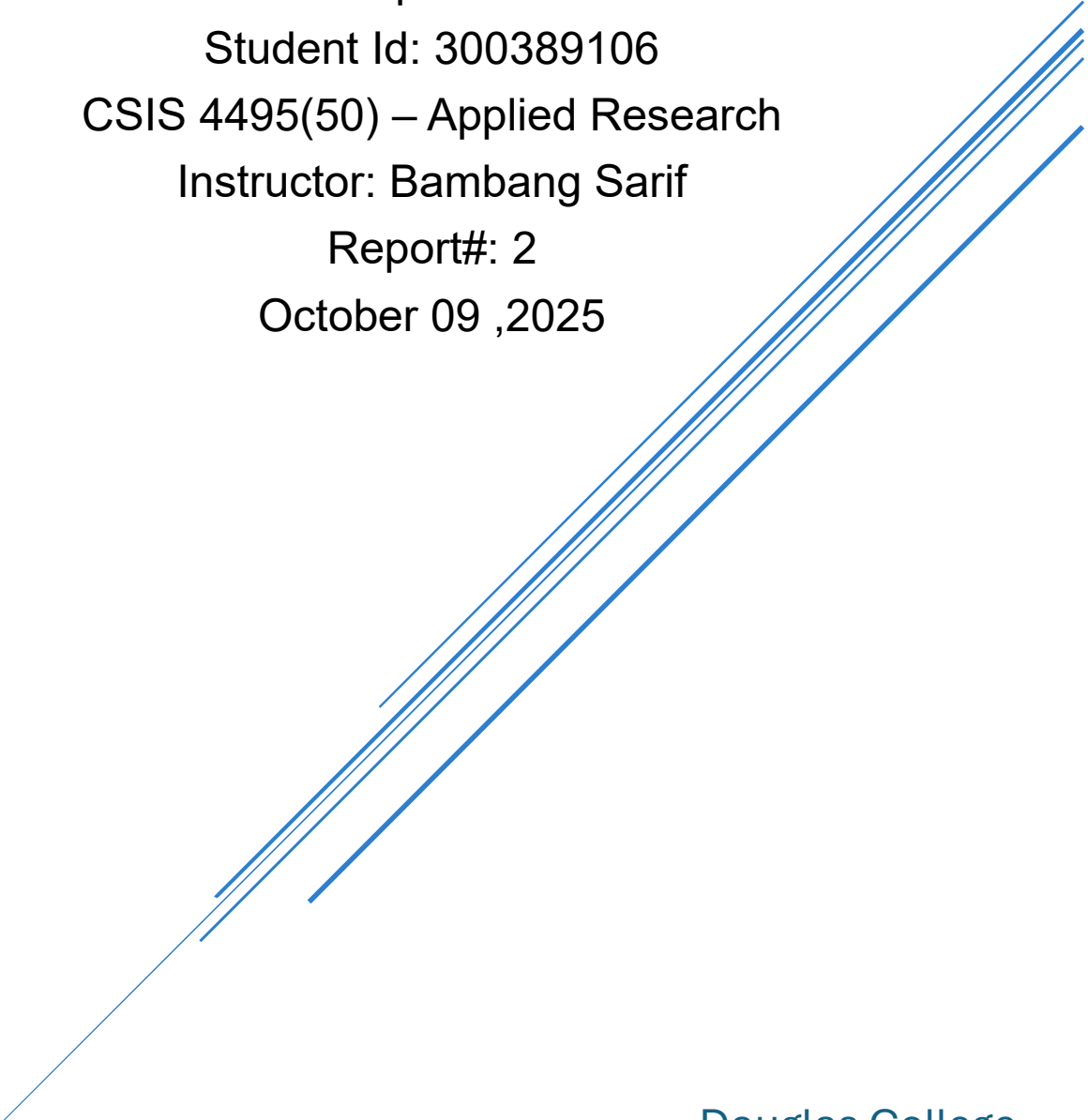
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CSIS 4495(50) – Applied Research

Instructor: Bambang Sarif

Report#: 2

October 09 ,2025

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Douglas College
CSIS 4495

Work Logs

Date	Number of Hours	Description of work done
Sep 27, 2025	2	Reviewed previous sprint tasks and set up new Kanban columns for Sprint 2 (Resident vs Volunteer dashboards).
Sep 30, 2025	3	Implemented _layout.tsx and navigation tabs for Resident and Volunteer dashboards.
Oct 1, 2025	2.5	Finished meeting with Harleen and discussed on topics, but moved on with the emergency response app.
Oct 3, 25	2	Created User Story and Task for Dynamic Dashboards (Resident vs Volunteer); updated project documentation.
Oct 7, 25	4	Developed dynamic dashboard layouts for Resident and Volunteer roles; added incident cards and status filters.
Oct 8, 25	6	Built Profile Screen to fetch user data via axios from /api/users/me; implemented logout and theme support.
Oct 9, 25	4.5	Connected backend to frontend for incident retrieval and submission; tested API calls using Expo and Postman. Prepared Report 2 and updated repo Documents/ with new screenshots and work logs.

Description of Work done

Since the previous report, major progress was made on both frontend and backend integration for the CERA application.

- Frontend: Built the complete login/register flow with Expo Router and secured token storage using AsyncStorage. Implemented role-based dashboards for Residents and Volunteers, each with custom tabs and actions (e.g., report incident vs accept/decline assignment).
- Backend: Integrated auth APIs (/auth/login, /auth/register, /users/me) and tested incident routes (/incidents, /assignments).
- UI/UX: Added Profile Screen with theme-adaptive colors and navigation to update profile or logout.
- Documentation: Updated architecture and Kanban board to include Sprint 2 tasks and epics for Dashboard and Role Management.

Challenges included fixing Expo Router navigation errors (useRouter and tab layout mismatches) and handling authentication state after login/logout.

While user was creating account; phone number and location was being sent as null.

These were resolved by centralizing router logic and token checks on app launch.

Data fields error was solved by hitting the submit API after fetching the location.

Repo Check-In

Checked-in folders and files since Report 1:

- F2025_4495_050_HKa855/app/Mobile/cera-mobile/tabs/profile/_layout.tsx; layout of Profile tab.
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/tabs/profile/history.tsx; to check the history of user or past volunteer work done (just dummy screen yet).
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/tabs/profile/index.tsx; main profile tab that shows- history, notifications, tasks and about the user.
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/tabs/profile/notifications.tsx; screen to show the notifications settings (just dummy screen yet).
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/tabs/profile/tasks.tsx; this screen will show the user assigned task or task that can be picked up (just dummy screen yet).
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/(tabs)/_layout.tsx; to show the tabs- Home, Tasks, Report and Profile so far.
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/(tabs)/home.tsx to show role-based dashboard screens.
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/(tabs)/report.tsx this will show any incident reported near user's location.
- F2025_4495_050_HKa855/app/Mobile/cera-mobile/(tabs)/tasks.tsx user will be able to see their active or assigned tasks here.
- F2025_4495_050_HKa855/app/components/ThemedText.tsx & ThemedView.tsx
- F2025_4495_050_HKa855/backend/routes/incident.js incidents schema updates

- F2025_4495_050_HKa855/backend/models/Incidents.js and API integration updates
- F2025_4495_050_HKa855/backend/server.js and API integration and other incident regarding updates
- F2025_4495_050_HKa855/Documents/Project_DSa855_Report2.pdf (this report)

Kanban Board

- **Epic 2:** Basic App & Backend Skeleton → Task completed (*Frontend breaking on initial start, Rebase main branch with bug fixes*).
- **Epic 2:** Basic App & Backend Skeleton → Tasks in progress (*Show user Dashboard based on the role*).
- **Epic 3:** Registration & Login → Tasks in progress (Role-Based Access).
- **Epic 2:** Basic App & Backend Skeleton → Tasks in Review/Test (*Fix the inputs posting- Phone number & Location*).

Images from GitHub Project

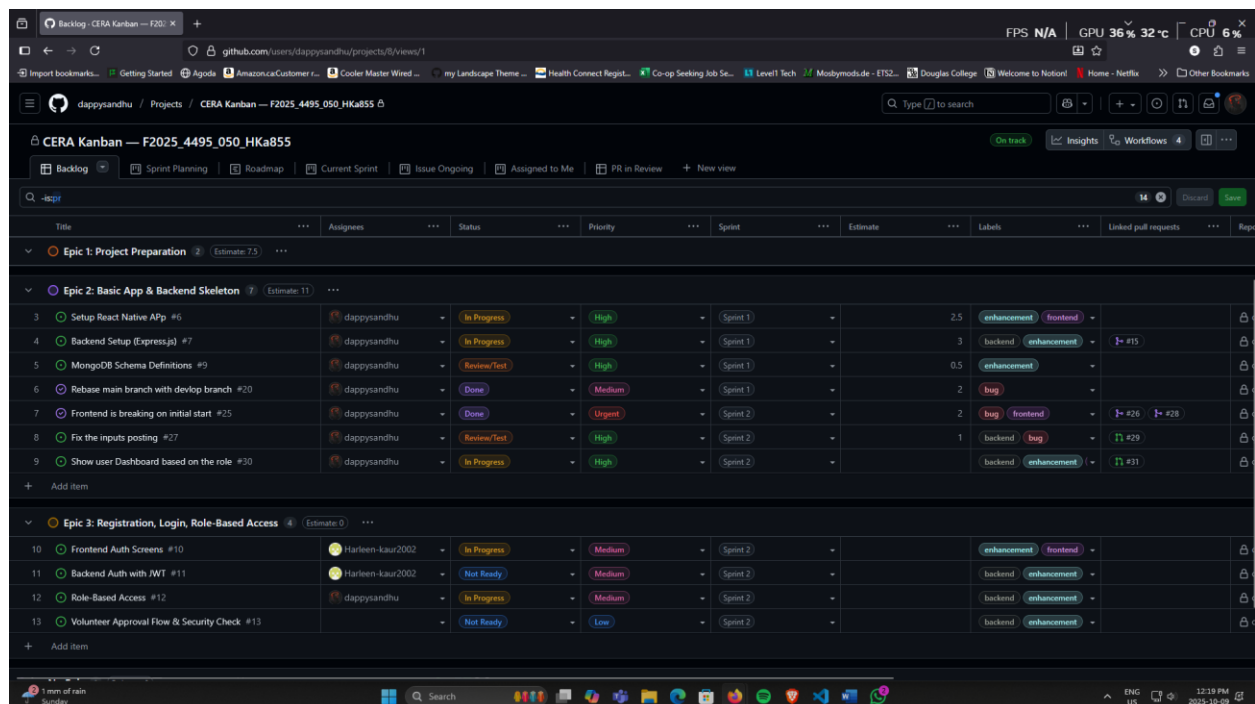


Figure 1: Kanban Board showing tasks in Backlog, In Progress, and Done.

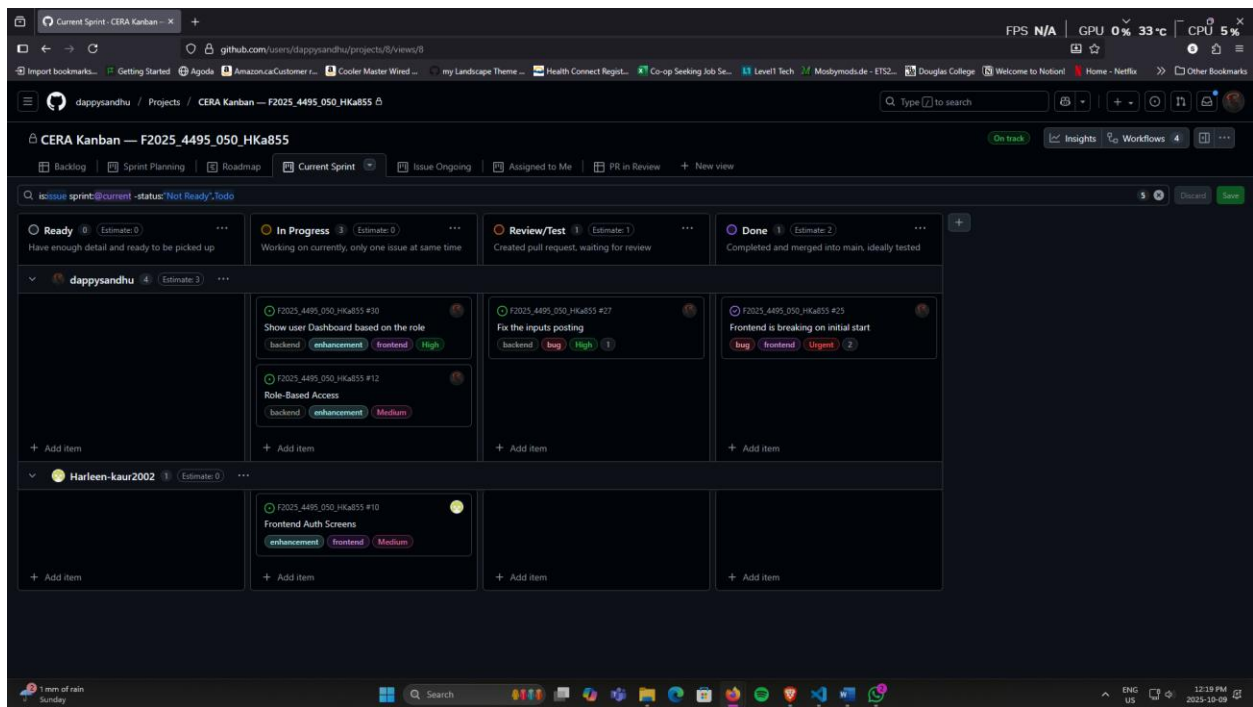


Figure 2: Current Ongoing Sprint.

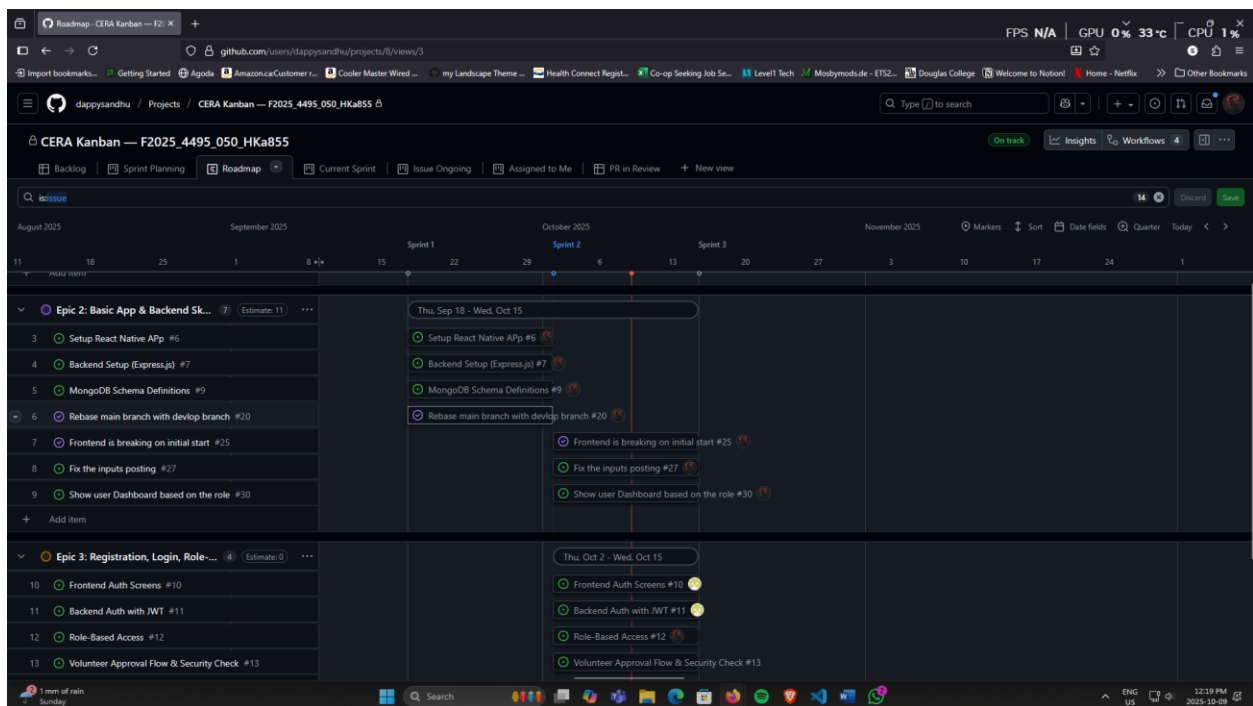
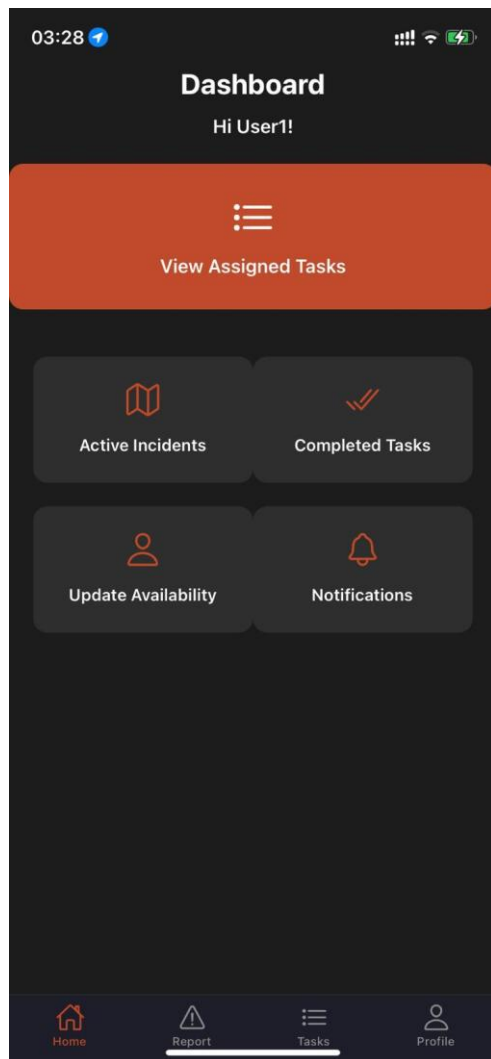


Figure 3: Project Roadmap.

Screens Implemented according to above tasks:



4(Dashboard)

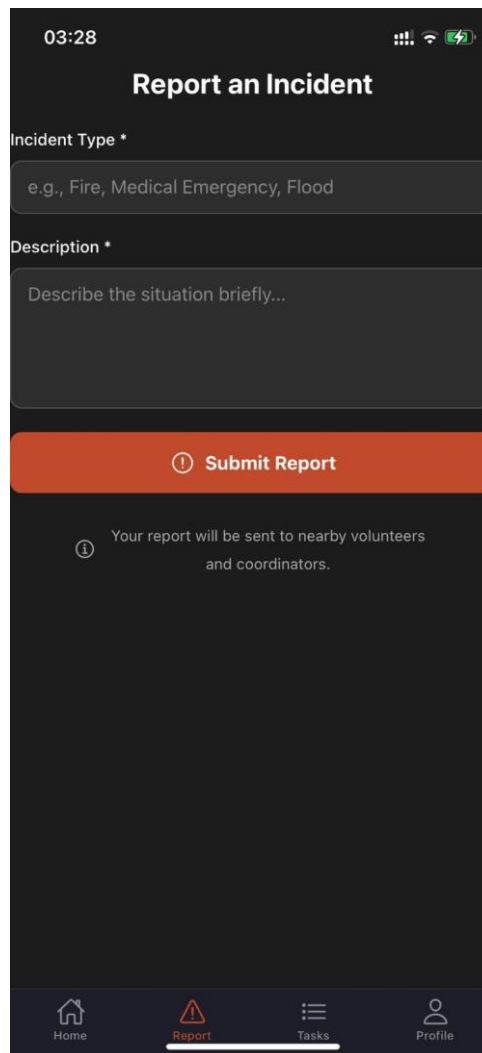


Figure 5(Reporting)

Figure

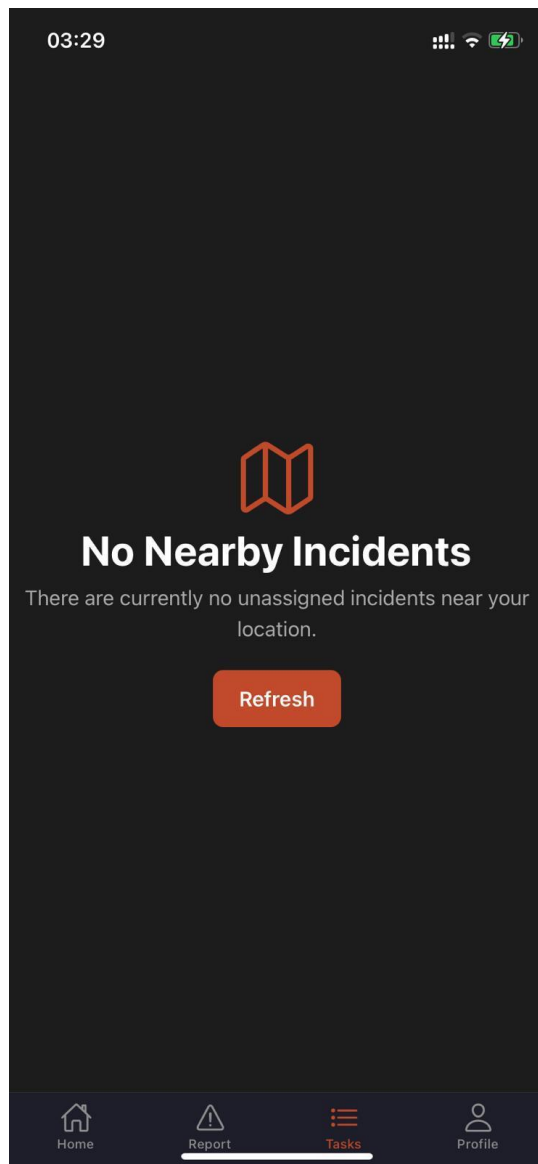


Figure 6(Incidents nearby)

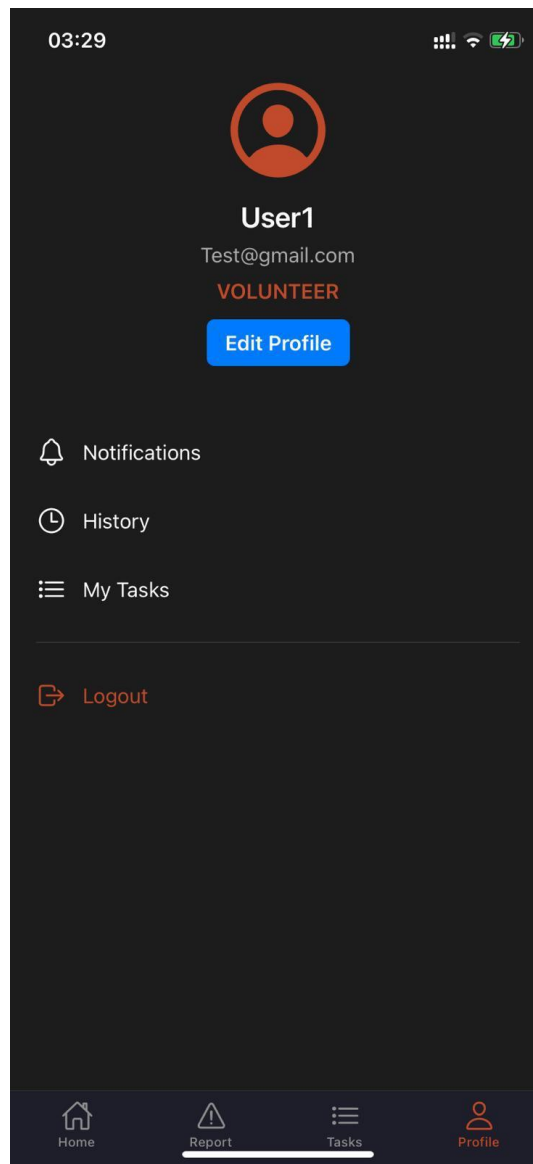


Figure 7(Profile)

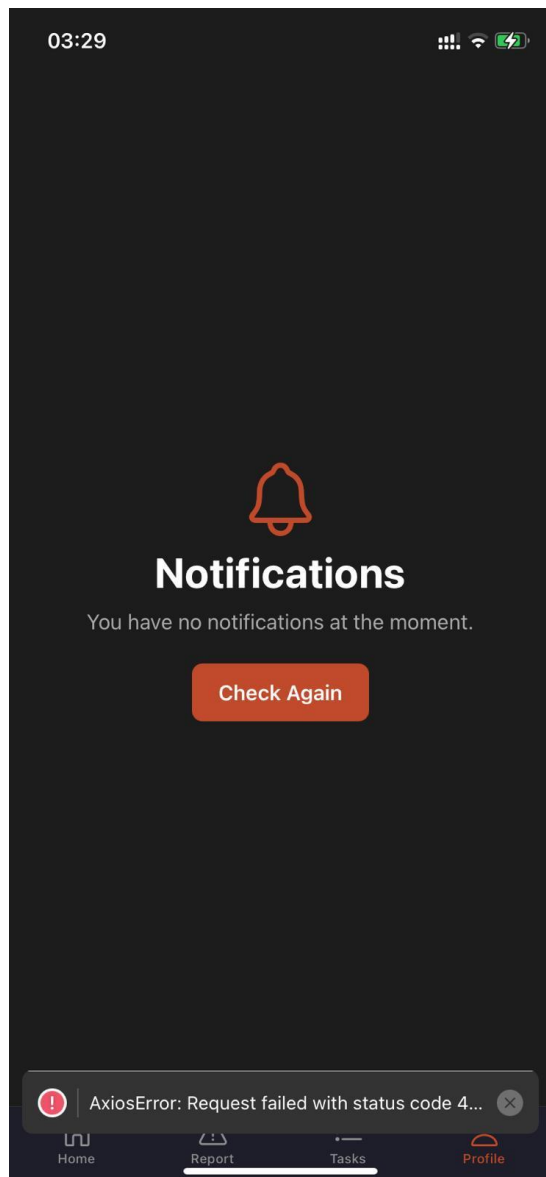


Figure 7.1(Notification from Profile)

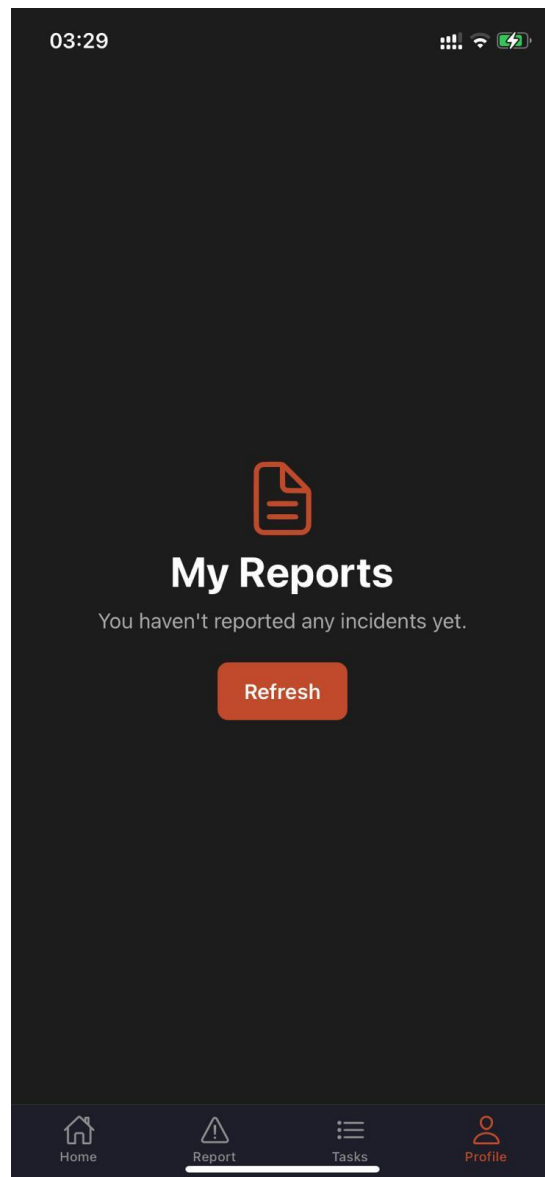


Figure 7.2(User's reports)

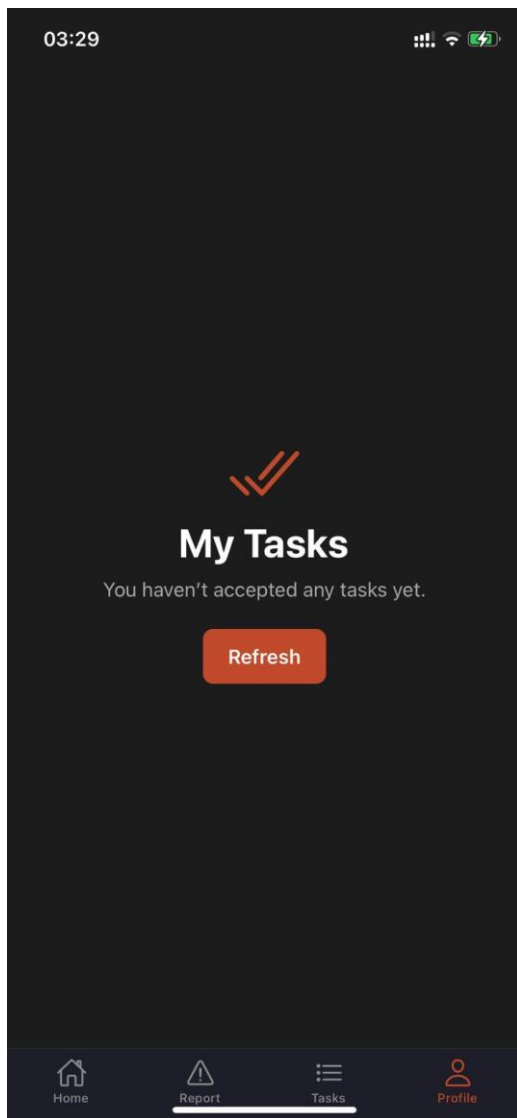


Figure 7.3(Tasks accepted by user)